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CONNECTOR

Abstract

A connector includes an electrically-conductive plug contact, a bottom insulator and an electrically-conductive inner contact, the plug contact having a tubular portion extending along a fitting axis with a recessed portion in an interior thereof and a flange extending from an end portion of the tubular portion, the bottom insulator having a support surface and a projection projecting from the support surface along the fitting axis and press-fitted in the recessed portion, the electrically-conductive inner contact being disposed on the support surface, the inner contact including a retention portion retained in the recessed portion and contacting an inner surface of the recessed portion and an elastic portion joined to the retention portion to face a back surface of the flange and having a cantilever shape elastically deformable in a direction along the fitting axis.

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Background/Summary

BACKGROUND OF THE INVENTION

[0001] The present invention relates to a connector, particularly to a connector connected to a sheet type connection object having a conductor exposed on at least one surface of the connection object.

[0002] In recent years, attention has been drawn to so-called smart clothes that can obtain user's biological data such as the heart rate and the body temperature only by being worn by the user. Such smart clothes have an electrode disposed at a measurement site, and when a wearable device serving as a measurement device is electrically connected to the electrode, biological data can be transmitted to the wearable device.

[0003] The electrode and the wearable device can be interconnected by, for instance, use of a connector connected to a conductor drawn from the electrode.

[0004] As a connector of this type, for example, JP 2018-129244 A discloses a connector shown in FIG. **26**. This connector includes a housing **2** and a base member **3** that are separately disposed on the opposite sides of a flexible substrate **1** to sandwich the flexible substrate **1**. Tubular portions **4**A of contacts **4** are passed through contact through-holes **2**A of the housing **2**, and flanges **4**B of the contacts **4** are sandwiched between the housing **2** and conductors **1**A exposed on the front surface of the flexible substrate **1**.

[0005] In this state, when the base member **3** is pushed toward the housing **2**, as shown in FIG. **27**, a projection **3**A of the base member **3** is inserted into a projection accommodating portion **4**C of the contact **4** with the flexible substrate **1** being sandwiched therebetween, and the inner surface of the projection accommodating portion **4**C makes contact with the conductor **1**A with a predetermined contact force, whereby the contact **4** is electrically connected to the conductor **1**A.

[0006] In addition, housing fixing posts **3**B formed to project on the base member **3** are press-fitted into post accommodating portions **2**B of the housing **2** as shown in FIG. **26**, so that the housing **2** and the base member **3** are fixed to each other.

[0007] When fitted with the connector disclosed in JP 2018-129244 A, a wearable device can be connected to electrodes formed of the conductors.

[0008] However, when the conductors **1**A are exposed only on the back surface of the flexible substrate **1**, the connector of JP 2018-129244 A is useless for electrically connecting the conductors **1**A to the contacts **4**, disadvantageously.

SUMMARY OF THE INVENTION

[0009] The present invention has been made to solve the foregoing problem and aims at providing a connector that enables an electrical connection of a contact to a conductor of a connection object regardless of whether the conductor is exposed on the front surface or the back surface of the connection object.

[0010] A connector according to the present invention comprises: [0011] a plug contact having electric conductivity and including a tubular portion and a flange, the tubular portion extending along a fitting axis and being provided in its interior with a recessed portion, and the flange extending from an end portion of the tubular portion in a direction orthogonal to the fitting axis; [0012] a bottom insulator including a support surface and a projection, the support surface extending in the direction orthogonal to the fitting axis, and the projection projecting from the support surface along the fitting axis and being press-fitted in the recessed portion; and [0013] an inner contact having electric conductivity and being disposed on the support surface, part of the inner contact being inserted in the recessed portion, [0014] wherein the inner contact includes a retention portion and an elastic portion, the retention portion being retained in the recessed portion and making contact with an inner surface of the recessed portion to be electrically connected to the plug contact, and the elastic portion being joined to the retention portion to face a back surface of the flange and having a cantilever shape that is elastically deformable in a direction along the fitting axis, [0015] when the projection is press-fitted in the recessed portion, the plug contact is fixed to the bottom insulator, and [0016] part of a connection object of sheet shape having a conductor exposed on at least one surface of the connection object is sandwiched between the back surface of the flange of the plug contact and the elastic portion of the inner contact in a direction along the fitting axis, the back surface of the flange makes contact with a front surface of the connection object, and the elastic portion makes contact with a back surface of the connection object, whereby the plug contact is electrically connected to the conductor directly when the conductor is exposed on the front surface of the connection object, and the plug contact is electrically connected to the conductor via the inner contact when the conductor is exposed on the back surface of the connection object.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0017] FIG. **1** is a perspective view of a connector according to an embodiment, when viewed from an obliquely upper position.

[0018] FIG. **2** is an assembly view of the connector according to the embodiment.

[0019] FIG. **3** is a perspective view showing a top insulator used in the connector of the embodiment.

[0020] FIG. **4** is a perspective view showing a plug contact used in the connector of the embodiment.

[0021] FIG. **5** is a cross-sectional view showing the plug contact used in the connector of the embodiment.

[0022] FIG. **6** is a perspective view showing a bottom insulator used in the connector of the embodiment.

[0023] FIG. **7** is a perspective view showing a projection of the bottom insulator used in the connector of the embodiment.

[0024] FIG. **8** is a plan view showing the projection of the bottom insulator used in the connector of the embodiment.

[0025] FIG. **9** is a perspective view of an inner contact used in the connector of the embodiment, when viewed obliquely from the front.

[0026] FIG. **10** is a perspective view of the inner contact used in the connector of the embodiment, when viewed obliquely from the rear.

[0027] FIG. **11** is a front view showing the inner contact used in the connector of the embodiment.

[0028] FIG. **12** is a perspective view of a connection object to be connected to the connector of the embodiment, when viewed from an obliquely upper position.

[0029] FIG. **13** is a perspective view of the connection object to be connected to the connector of the embodiment, when viewed from an obliquely lower position.

[0030] FIG. **14** is a perspective view showing a reinforcement sheet used in the connector of the embodiment.

[0031] FIG. **15** is a cross-sectional front view showing the state where the inner contact is aligned with the bottom insulator in the embodiment.

[0032] FIG. **16** is a perspective view showing the state where the inner contact is disposed in the bottom insulator in the embodiment.

[0033] FIG. **17** is a cross-sectional front view showing the state where the inner contact is disposed in the bottom insulator in the embodiment.

[0034] FIG. **18** is a plan view showing the state where the inner contact is disposed in the bottom insulator in the embodiment.

[0035] FIG. **19** is a cross-sectional front view showing the state where the connection object is disposed above the bottom insulator and the plug contact is aligned with the bottom insulator in the embodiment.
[0036] FIG. **20** is a cross-sectional front view showing the state where the projection of the bottom insulator is in the process of being press-fitted into the plug contact in the embodiment.
[0037] FIG. **21** is a cross-sectional front view showing a press-fitting completion state of the projection of the bottom insulator with respect to the plug contact in the embodiment.
[0038] FIG. **22** is a cross-sectional side view showing the press-fitting completion state of the projection of the bottom insulator with respect to the plug contact in the embodiment.
[0039] FIG. **23** is a perspective view of the connector of the embodiment connected to the connection object, when viewed from an obliquely lower position.
[0040] FIG. **24** is a cross-sectional front view partially showing the state of the interior of the plug contact in the connector of the embodiment connected to the connection object.
[0041] FIG. **25** is a cross-sectional side view partially showing the state of the interior of the plug contact in the connector of the embodiment connected to the connection object.
[0042] FIG. **26** is an exploded perspective view showing a conventional connector.
[0043] FIG. **27** is a partial cross-sectional view showing the conventional connector.
DETAILED DESCRIPTION OF THE INVENTION
[0044] An embodiment of the present invention is described below based on the accompanying drawings.
[0045] FIG. **1** shows a connector **11** according to the embodiment. The connector **11** is used as, for instance, a garment-side connector for fitting a wearable device and has a housing **12** made of an insulating material. In the housing **12**, four plug contacts **13** are retained, and a reinforcement sheet **14** and a sheet type conductive member **15** being superposed on each other are retained by the housing **12**. The sheet type conductive member **15** constitutes a connection object to which the contact **11** is connected.
[0046] The four plug contacts **13** are aligned in two rows parallel to each other and disposed to project perpendicularly with respect to the sheet type conductive member **15**.
[0047] For convenience, the reinforcement sheet **14** and the sheet type conductive member **15** are defined as extending along an XY plane, the alignment direction of the four plug contacts **13** is referred to as “Y direction,” and the direction in which the four plug contacts **13** project is referred to as “+Z direction.” The Z direction is a fitting direction in which the connector **11** is fitted to a counter connector.
[0048] FIG. **2** shows an exploded perspective view of the connector **11**. The connector **11** includes a top insulator **16** and a bottom insulator **17**, and these top and bottom insulators **16** and **17** constitute the housing **12**.
[0049] The four plug contacts **13** are disposed on the −Z direction side of the top insulator **16**, the reinforcement sheet **14** is disposed on the −Z direction side of the plug contacts **13**, and the sheet type conductive member **15** is disposed on the −Z direction side of the reinforcement sheet **14**. Further, four inner contacts **18** are disposed on the −Z direction side of the sheet type conductive member **15**, and the bottom insulator **17** is disposed on the −Z direction side of the inner contacts **18**. The four inner contacts **18** separately correspond to the four plug contacts **13**.
[0050] As shown in FIG. **3**, the top insulator **16** includes a recessed portion **16**A opening in the +Z direction and four contact through-holes **16**B formed within the recessed portion **16**A. The recessed portion **16**A constitutes a counter connector accommodating portion in which part of a counter connector (not shown) is to be accommodated, and the four contact through-holes **16**B separately correspond to the four plug contacts **13**. In addition, a plurality of bosses **16**C protruding in the −Z direction are formed on a surface, facing in the −Z direction, of the top insulator **16**.
[0051] The four plug contacts **13** are each made of a conductive material such as metal and are to be connected to corresponding contacts of a counter connector (not shown) when part of the

counter connector is accommodated in the recessed portion **16**A of the top insulator **16**.
[0052] As shown in FIG. **4**, the plug contact **13** includes a tubular portion **13**A of cylindrical shape extending in the Z direction along a fitting axis C, and a flange **13**B extending from a −Z directional end portion of the tubular portion **13**A along an XY plane.
[0053] As shown in FIG. **5**, the tubular portion **13**A is provided in its interior with a recessed portion **13**C opening in the −Z direction. The recessed portion **13**C has an inside diameter D**1** in the vicinity of the −Z directional end portion of the tubular portion **13**A.
[0054] It should be noted that the fitting axis C is an axis passing the center of the tubular portion **13**A and extending in the fitting direction between the connector **11** and a counter connector.
[0055] While the tubular portion **13**A has a cylindrical shape, the cross-sectional shape thereof is not limited to a circular shape, and the tubular portion **13**A may have any of various cross-sectional shapes such as ellipses and polygons as long as the tubular portion **13**A is provided in its interior with the recessed portion **13**C.
[0056] All the four plug contacts **13** can be used as terminals for transmitting electric signals.
[0057] As shown in FIG. **6**, the bottom insulator **17** includes a flat plate portion **17**A extending along an XY plane, and a surface of the flat plate portion **17**A facing in the +Z direction forms a flat placement surface S used to place the sheet type conductive member **15**. The placement surface S of the flat plate portion **17**A is provided with four recessed shape portions **17**B opening in the +Z direction. The four recessed shape portions **17**B separately correspond to the four plug contacts **13**. The four recessed shape portions **17**B are separately provided with four projections **17**C projecting from the centers of the recessed shape portions **17**B in the +Z direction.
[0058] In addition, the flat plate portion **17**A is provided with a plurality of through-holes **17**D separately corresponding to the plurality of bosses **16**C of the top insulator **16**.
[0059] As shown in FIGS. **7** and **8**, the bottom surface of the recessed shape portion **17**B of the bottom insulator **17** forms a flat support surface **17**E extending along an XY plane and facing in the +Z direction.
[0060] The projection **17**C projects from the support surface **17**E in the +Z direction along the fitting axis C and has a prismatic shape having a substantially hexagonal cross section. The projection **17**C is provided, at its lateral surface along the circumference, with six press-fit portions **17**F protruding in the radiation direction with the fitting axis C as the center and extending along the fitting axis C.
[0061] The projection **17**C is also provided at its lateral surface with a pair of insertion grooves **17**G separately extending along the fitting axis C in different positions from the press-fit portions **17**F. These insertion grooves **17**G are used to retain the inner contact **18**. One insertion groove **17**G opens in the +Y direction, and the other insertion groove **17**G opens in the −Y direction.
[0062] A distance D**2** between a pair of press-fit portions **17**F provided in the opposite directions across the fitting axis C is set to be slightly larger than the inside diameter D**1** of the recessed portion **13**C of the plug contact **13** shown in FIG. **5**.
[0063] Thus, when the connector **11** is assembled, the plug contact **13** can be fixed to the bottom insulator **17** by press-fitting the projection **17**C of the bottom insulator **17** into the recessed portion **13**C of the plug contact **13**.
[0064] Besides, bottoms of the pair of insertion grooves **17**G formed at the lateral surface of the projection **17**C are separated from each other by a distance Y**1** along the Y direction.
[0065] As shown in FIGS. **9** to **10**, the inner contact **18** is formed of a single metal sheet having electric conductivity and being bent, and includes a base portion **18**A extending along an XY plane, a retention portion **18**B joined to the base portion **18**A, and a pair of elastic portions **18**C joined to the base portion **18**A. When the connector **11** has been assembled, the base portion **18**A is disposed on the support surface **17**E of the corresponding recessed shape portion **17**B of the bottom insulator **17**, and the retention portion **18**B is inserted in the recessed portion **13**C of the corresponding plug contact **13**.

[0066] The base portion **18**A is a flat plate member extending in a substantially semicircular form along an XY plane while curving around the fitting axis C. The retention portion **18**B is joined to the opposite ends, in the Y direction, of the base portion **18**A, and the pair of elastic portions **18**C are also joined to the opposite ends, in the Y direction, of the base portion **18**A.
[0067] The retention portion **18**B is a flat member extending along a YZ plane and cut out in a substantially U shape so as to surround the projection **17**C of the bottom insulator **17**. The retention portion **18**B is provided, at its +Y and −Y directional ends at a predetermined height in the Z direction, with a pair of first contact portions P**1** that are spaced from each other in the Y direction and face in the opposite directions from each other.
[0068] The retention portion **18**B also has a pair of pressing portions **18**D on the −Z direction side of the pair of first contact portions P**1**, the pair of pressing portions **18**D facing each other across the fitting axis C and protruding toward the fitting axis C. When the connector **11** has been assembled, the projection **17**C of the bottom insulator **17** is press-fitted between the pair of pressing portions **18**D with the pair of pressing portions **18**D being inserted in the pair of insertion grooves **17**G of the projection **17**C.
[0069] The pair of elastic portions **18**C each have a cantilever shape that extends in the +X direction from the corresponding Y directional end portion of the base portion **18**A, rises in the +Z direction, and is bent to return to the +Z directional side of the corresponding Y directional end portion of the base portion **18**A. Because of the shape as above, the pair of elastic portions **18**C are each configured to be elastically deformable in the Z direction. The pair of elastic portions **18**C are respectively provided at their ends with a pair of second contact portions P**2** facing in the +Z direction.
[0070] As shown in FIG. **11**, a distance Y**2** in the Y direction between the pair of first contact portions P**1** is set to be slightly larger than the inside diameter D**1** of the part of the recessed portion **13**C with which the pair of first contact portions P**1** make contact when the retention portion **18**B is inserted into the recessed portion **13**C of the plug contact **13**.
[0071] Accordingly, the pair of first contact portions P**1** make contact with an inner surface of the recessed portion **13**C with predetermined contact pressure while being elastically displaced in a direction toward the fitting axis C when the retention portion **18**B of the inner contact **18** is inserted into the recessed portion **13**C of the plug contact **13**.
[0072] A distance Y**3** in the Y direction between the pair of pressing portions **18**D facing each other is set to be slightly smaller than the distance Y**1** in the Y direction between the bottoms of the pair of insertion grooves **17**G formed at the projection **17**C of the bottom insulator **17** shown in FIG. **8**.
[0073] Accordingly, when the connector **11** is assembled, the inner contact **18** is pushed toward the bottom insulator **17** while the pair of pressing portions **18**D of the inner contact **18** is inserted in the pair of insertion grooves **17**G of the projection **17**C of the bottom insulator **17**, whereby the projection **17**C is press-fitted between the pair of pressing portions **18**D. Hence, the inner contact **18** can be fixed to the bottom insulator **17**.
[0074] The sheet type conductive member **15** has a multilayer structure in which a plurality of wiring layers each formed from a conductor and a plurality of insulating layers are laminated.
[0075] As shown in FIG. **12**, four contact arrangement regions **15**A for arranging the four plug contacts **13** are defined on the front surface, facing in the +Z direction, of the sheet type conductive member **15**. Each contact arrangement region **15**A is provided at its central part with a circular opening portion **15**B penetrating the sheet type conductive member **15** in the Z direction, and a wiring layer **15**C is exposed toward the +Z direction around the opening portion **15**B so as to surround the opening portion **15**B. An insulating layer **15**D is exposed in the other region than the four contact arrangement regions **15**A.
[0076] Since the opening portions **15**B penetrate the sheet type conductive member **15** in the Z direction, as shown in FIG. **13**, the opening portions **15**B can be seen also on the back surface, facing in the −Z direction, of the sheet type conductive member **15** at positions corresponding to

the four contact arrangement regions **15**A.

[0077] On the back surface, facing in the −Z direction, of the sheet type conductive member **15**, a wiring layer **15**E is exposed toward the −Z direction around the respective opening portions **15**B formed at the positions corresponding to the four contact arrangement regions **15**A so as to surround the respective opening portions **15**B, and an insulating layer **15**F is exposed in the remaining region.

[0078] In addition, as shown in FIGS. **12** and **13**, a plurality of through-holes **15**G separately corresponding to the plurality of bosses **16**C of the top insulator **16** are formed in a peripheral portion of the sheet type conductive member **15**.

[0079] As shown in FIG. **14**, the reinforcement sheet **14** is used to reinforce a mounting object (not shown) such as a garment on which the connector **11** is to be mounted. The reinforcement sheet **14** is made of an insulating material and provided at its center with an opening portion **14**A. Further, a plurality of cutouts **14**B separately corresponding to the plurality of bosses **16**C of the top insulator **16** are formed along the circumference of the opening portion **14**A of the reinforcement sheet **14**.

[0080] The four contact through-holes **16**B of the top insulator **16**, the four plug contacts **13**, the four contact arrangement regions **15**A of the sheet type conductive member **15**, the four inner contacts **18**, and the four recessed shape portions **17**B of the bottom insulator **17** are arranged to align with each other in the Z direction.

[0081] In addition, the bosses **16**C of the top insulator **16**, the cutouts **14**B of the reinforcement sheet **14**, the through-holes **15**G of the sheet type conductive member **15**, and the through-holes **17**D of the bottom insulator **17** are arranged to align with each other in the Z direction.

[0082] When the connector **11** is assembled, first, the inner contact **18** is positioned with respect to the corresponding recessed shape portion **17**B of the bottom insulator **17** on the +Z direction side thereof, as shown in FIG. **15**. At this time, the base portion **18**A and the pair of elastic portions **18**C of the inner contact **18** are situated on the +Z direction side of the corresponding recessed shape portion **17**B of the bottom insulator **17**, and the retention portion **18**B is situated on the +Z direction side of the projection **17**C projecting from the recessed shape portion **17**B.

[0083] Next, the inner contact **18** is pushed in the −Z direction toward the recessed shape portion **17**B of the bottom insulator **17**, while the projection **17**C of the bottom insulator **17** is inserted to the inside of the substantially U-shaped retention portion **18**B of the inner contact **18**, as shown in FIG. **16**. At this time, the inner contact **18** is pushed with the pair of pressing portions **18**D projecting toward the inside of the retention portion **18**B of the inner contact **18** being inserted in the pair of insertion grooves **17**G of the projection **17**C of the bottom insulator **17**, as shown in FIG. **17**.

[0084] Since the distance Y**3** in the Y direction between the pair of pressing portions **18**D of the inner contact **18** shown in FIG. **11** is set to be slightly smaller than the distance Y**1** in the Y direction between the bottoms of the pair of insertion grooves **17**G of the bottom insulator **17** shown in FIG. **8**, when the inner contact **18** is pushed toward the bottom insulator **17**, the projection **17**C is press-fitted between the pair of pressing portions **18**D, so that the inner contact **18** is fixed to the bottom insulator **17**.

[0085] As shown in FIG. **17**, the inner contact **18** is pushed toward the bottom insulator **17** until the base portion **18**A and the pair of elastic portions **18**C of the inner contact **18** come into contact with the support surface **17**E of the recessed shape portion **17**B of the bottom insulator **17**.

[0086] The elastic portions **18**C of the inner contact **18** have a Z directional height larger than the depth dimension of the recessed shape portion **17**B of the bottom insulator **17**, and accordingly, the pair of second contact portions P**2** disposed at the ends of the pair of elastic portions **18**C protrude in the +Z direction from the recessed shape portion **17**B of the bottom insulator **17** and situated on the +Z direction side of the placement surface S of the bottom insulator **17**.

[0087] Thus, the inner contact **18** is accommodated in the recessed shape portion **17**B of the bottom insulator **17** and supported on the support surface **17**E as shown in FIG. **18**.

[0088] Next, the sheet type conductive member **15** is disposed above the bottom insulator **17** as shown in FIG. **19**. At this time, the projection **17**C of the bottom insulator **17** and the retention portion **18**B of the inner contact **18** fixed to the projection **17**C are inserted into the corresponding opening portion **15**B of the sheet type conductive member **15**, and the contact arrangement region **15**A is situated on the +Z direction side of the corresponding recessed shape portion **17**B of the bottom insulator **17**.
[0089] However, since the pair of second contact portions P**2** of the inner contact **18** protrude in the +Z direction from the recessed shape portion **17**B of the bottom insulator **17** and the sheet type conductive member **15** is situated on the pair of second contact portions P**2**, the contact arrangement region **15**A of the sheet type conductive member **15** is in the state of floating on the +Z direction side of the placement surface S of the bottom insulator **17**.
[0090] Further, the corresponding plug contact **13** is positioned on the +Z direction side of the sheet type conductive member **15**. At this time, the recessed portion **13**C of the plug contact **13** is situated on the +Z direction side of the retention portion **18**B of the inner contact **18** fixed to the corresponding projection **17**C of the bottom insulator **17**.
[0091] Since the distance Y**2** in the Y direction between the pair of first contact portions P**1** of the inner contact **18** shown in FIG. **11** is set to be slightly larger than the inside diameter D**1** of the recessed portion **13**C of the plug contact **13** shown in FIG. **5**, the −Z directional end of the recessed portion **13**C of the plug contact **13** is caught on the pair of first contact portions P**1** as shown in FIG. **19**.
[0092] In this state, the plug contact **13** is pushed toward the bottom insulator **17** relatively in the −Z direction. As a consequence, as shown in FIG. **20**, the plug contact **13** is moved in the −Z direction while compressing and deforming the retention portion **18**B of the inner contact **18** in the Y direction, the retention portion **18**B of the inner contact **18** and the projection **17**C of the bottom insulator **17** are inserted into the recessed portion **13**C of the plug contact **13**, and the back surface of the flange **13**B of the plug contact **13** makes contact with the front surface, on the +Z direction side, of the sheet type conductive member **15**.
[0093] When the plug contact **13** is further pushed in the −Z direction, the sheet type conductive member **15** is pushed in the −Z direction by the flange **13**B of the plug contact **13**, and the plug contact **13** and the sheet type conductive member **15** are moved in the −Z direction while compressing the pair of elastic portions **18**C in the Z direction via the pair of second contact portions P**2**, whereby the sheet type conductive member **15** makes contact with the placement surface S of the bottom insulator **17**, as shown in FIGS. **22** and **23**.
[0094] Since the distance D**2** between the pair of press-fit portions **17**F of the projection **17**C provided in the opposite directions across the fitting axis C is set to be slightly larger than the inside diameter D**1** of the recessed portion **13**C of the plug contact **13** as described above, when the plug contact **13** is pushed toward the bottom insulator **17** as shown in FIGS. **19** to **22**, the projection **17**C is press-fitted into the recessed portion **13**C, and the plug contact **13** is fixed to the bottom insulator **17**.
[0095] In this state, the reinforcement sheet **14** and the top insulator **16** are sequentially disposed with respect to the sheet type conductive member **15** from the +Z direction side. In this process, the tubular portions **13**A of the plug contacts **13** are correspondingly inserted into the four contact through holes **16**B of the top insulator **16** from the −Z direction, and the bosses **16**C of the top insulator **16** sequentially penetrate the cutouts **14**B of the reinforcement sheet **14**, the through-holes **15**G of the sheet type conductive member **15**, and the through-holes **17**D of the bottom insulator **17**. Then, as shown in FIG. **23**, the top insulator **16** and the bottom insulator **17** are fixed to each other through heat deformation of the tips of the bosses **16**C protruding on the −Z direction side of the bottom insulator **17**. Thus, the assembling operation of the connector **11** is completed.
[0096] In the thus-assembled connector **11**, the sheet type conductive member **15** is sandwiched between the flange **13**B of the plug contact **13** and the pair of elastic portions **18**C of the inner

contact **18** as shown in FIG. **24**. In addition, the wiring layer **15**C exposed on the front surface, facing in the +Z direction, of the sheet type conductive member **15** faces the back surface of the flange **13**B of the plug contact **13**, while the wiring layer **15**E exposed on the back surface, facing in the −Z direction, of the sheet type conductive member **15** faces the pair of second contact portions P**2** of the pair of elastic portions **18**C of the inner contact **18**.

[0097] Consequently, the back surface of the flange **13**B of the plug contact **13** makes contact with the wiring layer **15**C of the sheet type conductive member **15** with predetermined contact pressure, so that the plug contact **13** is electrically connected to the wiring layer **15**C. In addition, the pair of second contact portions P**2** of the pair of elastic portions **18**C of the inner contact **18** make contact with the wiring layer **15**E of the sheet type conductive member **15** with predetermined contact pressure, so that the inner contact **18** is electrically connected to the wiring layer **15**E.

[0098] When the retention portion **18**B of the inner contact **18** is inserted in the recessed portion **13**C of the plug contact **13**, and the pair of first contact portions P**1** formed at the retention portion **18**B make contact with the inner surface of the recessed portion **13**C of the plug contact **13** with predetermined contact pressure, the inner contact **18** is electrically connected to the plug contact **13**.

[0099] Therefore, the wiring layer **15**C exposed on the front surface of the sheet type conductive member **15** is electrically connected to the plug contact **13** directly, while the wiring layer **15**E exposed on the back surface of the sheet type conductive member **15** is electrically connected to the plug contact **13** via the inner contact **18**. That is, both the wiring layers **15**C and **15**E are connected to the plug contact **13**.

[0100] FIG. **25** is a cross-sectional view of the assembled connector **11** as seen from a direction different from that of FIG. **24**, i.e., from a lateral side.

[0101] Thus, in the connector **11**, the use of the inner contact **18** makes it possible to electrically connect both the wiring layer **15**C and the wiring layer **15**E respectively formed of conductors disposed on the front surface side and the back surface side of the sheet type conductive member **15** to one plug contact **13**.

[0102] Therefore, when the connector **11** is connected to a sheet type conductive member having a conductor exposed only on its front surface side, the plug contact **13** can be electrically connected to the conductor on the front surface side of the sheet type conductive member. On the other hand, when the connector **11** is connected to a sheet type conductive member having a conductor exposed only on its back surface side, the plug contact **13** can be electrically connected to the conductor on the back surface side of the sheet type conductive member.

[0103] Further, when the connector **11** is connected to a sheet type conductive member having conductors separately exposed on its front surface side and back surface side like the sheet type conductive member **15** in this embodiment, the plug contact **13** can be electrically connected to both the conductors on the front surface side and the back surface side of the sheet type conductive member. For example, assuming that a connection object is a sheet type conductive member having a multilayer structure in which conductors constituting shield layers are separately exposed on the front surface side and the back surface side and a conductor constituting a signal wiring layer is disposed between those shield layers so as to be insulated from both the shield layers, a shield effect is exhibited with respect to the signal wiring layer when the plug contact **13** connected to the shield layers on the front surface side and the back surface side is connected to a ground potential, thus enabling to carry out highly accurate signal transmission with reduced influence of external disturbances caused by, for example, electromagnetic waves.

[0104] It should be noted that in the above embodiment, it is not necessary to bend a part of the sheet type conductive member **15** at a substantially right angle to catch and bring it inside the recessed portion **13**C of the plug contact **13** like the connector of JP 2018-129244 A shown in FIG. **27**; the plug contact **13** can be electrically connected to the wiring layer **15**C, **15**E of the sheet type conductive member **15** with the sheet type conductive member **15** extending along an XY plane being sandwiched between the flange **13**B of the plug contact **13** and the pair of elastic portions **18**C of the inner contact **18**. Thus, disconnection of the wiring layer **15**C, **15**E of the sheet type conductive member **15** due to bending can be prevented from occurring, and in addition, the plug contact **13** can be electrically connected to a connection object that is hard to bend.

[0105] In the above embodiment, the projection **17**C of the bottom insulator **17** is press-fitted into the recessed portion **13**C of the tubular portion **13**A of the plug contact **13**, whereby the plug contact **13** is fixed to the bottom insulator **17**. This configuration makes it possible to electrically connect the plug contact **13** to the wiring layer **15**C, **15**E of the sheet type conductive member **15** without the use of the top insulator **16**.

[0106] In regard to fixation of the plug contact **13** to the bottom insulator **17**, the flange **13**B of the plug contact **13** may be held down with the top insulator **16** to fix the plug contact **13** to the bottom insulator **17** by passing the bosses **16**C of the top insulator **16** through the through-holes **17**D of the bottom insulator **17** and thermally deforming the tips of the bosses **16**C or bonding and fixing the top insulator **16** to the bottom insulator **17** with an adhesive.

[0107] However, it is difficult to arrange the bosses **16**C, which pass through the through-holes **17**D, in the vicinity of the plug contact **13** and also difficult to use an adhesive near the plug contact **13** while preventing spreading of the adhesive which may cause poor contact. Accordingly, the top insulator **16** and the bottom insulator **17** may warp due to a reaction force of a contact load in the fitting direction, so that a distance may be increased between the flange **13**B of the plug contact **13** and the pair of elastic portions **18**C of the inner contact **18**, resulting in impaired reliability of the electrical connection.

[0108] In contrast, in the above embodiment, the projection **17**C of the bottom insulator **17** is press-fitted into the recessed portion **13**C of the plug contact **13**, whereby the plug contact **13** is fixed to the bottom insulator **17**; therefore, the bottom insulator **17** is prevented from warping, thus ensuring the reliability of the electrical connection.

[0109] Aside from that, in the above embodiment, the base portion **18**A of the inner contact **18** is disposed on the flat support surface **17**E that is formed at the bottom surface of the recessed shape portion **17**B of the bottom insulator **17** and extends along an XY plane. Owing to this configuration, the inner contact **18** is stably supported inside the connector **11**, thus improving the reliability of the electrical connection of the plug contact **13** with the wiring layer **15**C, **15**E of the sheet type conductive member **15**.

[0110] While the inner contact **18** has the pair of elastic portions **18**C in the above embodiment, the invention is not limited thereto, and the inner contact **18** may have a single elastic portion **18**C or three or more elastic portions **18**C.

[0111] In the above embodiment, the plug contacts **13** disposed in the contact arrangement regions **15**A of the sheet type conductive member **15** are connected to both the wiring layer **15**C exposed on the front surface side of the sheet type conductive member **15** and the wiring layer **15**E exposed on the back surface side of the sheet type conductive member **15**; however, for instance, it is also possible to connect only the wiring layer **15**E exposed on the back surface side of the sheet type conductive member **15** to the plug contacts **13** disposed in the contact arrangement regions **15**A.

[0112] While the sheet type conductive member **15** used in the above embodiment has a multilayer structure, the invention is not limited thereto, and it suffices if a conductive member has a conductor exposed on at least one surface.

[0113] In addition, while two layers of the conductors, i.e., the wiring layer **15**C and the wiring layer **15**E of the sheet type conductive member **15** are connected to one plug contact **13** in the above embodiment, the invention is not limited thereto, and three or more layers of conductors may be connected to one plug contact **13**.

[0114] In addition, while the connector **11** in the above embodiment has the four plug contacts **13**, the invention is not limited to this number of the plug contacts **13**, and it suffices if the connector includes at least one plug contact **13** electrically connected to a conductor exposed on at least one surface of the sheet type conductive member **15**.

[0115] While the reinforcement sheet **14** is disposed between the bottom insulator **17** and the top insulator **16** in the above embodiment, the reinforcement sheet **14** may be omitted when it is not necessary to reinforce a mounting object such as a garment to which the connector **11** is to be attached.

Claims

1. A connector comprising: a plug contact having electric conductivity and including a tubular portion and a flange, the tubular portion extending along a fitting axis and being provided in its interior with a recessed portion, and the flange extending from an end portion of the tubular portion in a direction orthogonal to the fitting axis; a bottom insulator including a support surface and a projection, the support surface extending in the direction orthogonal to the fitting axis, and the projection projecting from the support surface along the fitting axis and being press-fitted in the recessed portion; and an inner contact having electric conductivity and being disposed on the support surface, part of the inner contact being inserted in the recessed portion, wherein the inner contact includes a retention portion and an elastic portion, the retention portion being retained in the recessed portion and making contact with an inner surface of the recessed portion to be electrically connected to the plug contact, and the elastic portion being joined to the retention portion to face a back surface of the flange and having a cantilever shape that is elastically deformable in a direction along the fitting axis, when the projection is press-fitted in the recessed portion, the plug contact is fixed to the bottom insulator, and part of a connection object of sheet shape having a conductor exposed on at least one surface of the connection object is sandwiched between the back surface of the flange of the plug contact and the elastic portion of the inner contact in a direction along the fitting axis, the back surface of the flange makes contact with a front surface of the connection object, and the elastic portion makes contact with a back surface of the connection object, whereby the plug contact is electrically connected to the conductor directly when the conductor is exposed on the front surface of the connection object, and the plug contact is electrically connected to the conductor via the inner contact when the conductor is exposed on the back surface of the connection object.

2. The connector according to claim 1, wherein the inner contact is formed of a metal sheet being bent and includes: a base portion of flat plate shape extending in the direction orthogonal to the fitting axis and being disposed on the support surface; the retention portion of flat plate shape being joined to the base portion and being cut out in a U shape so as to surround the projection along a plane including the fitting axis; and a pair of elastic portions each identical to the elastic portion, the pair of elastic portions being joined to the base portion and disposed at positions facing each other across the fitting axis.

3. The connector according to claim 2, wherein the retention portion has a pair of first contact portions that are spaced from each other in the direction orthogonal to the fitting axis and separately make contact with the inner surface of the recessed portion, and the pair of elastic portions have a pair of second contact portions disposed at ends of the pair of elastic portions, respectively.

4. The connector according to claim 3, wherein the retention portion includes a pair of pressing portions facing each other across the fitting axis and separately protruding in directions toward the fitting axis, and the projection is press-fitted between the pair of pressing portions, whereby the inner contact is retained by the bottom insulator.

5. The connector according to claim 4, wherein the projection includes a pair of insertion grooves in which the pair of pressing portions are inserted.

6. The connector according to claim 1, wherein the projection includes a plurality of press-fit portions that protrude in a radiation direction with the fitting axis as a center from a lateral surface of the projection extending along a circumference of the projection and that extend along the fitting axis, and the plurality of press-fit portions are press-fitted in the recessed portion.

7. The connector according to claim 1, wherein the bottom insulator includes a placement surface which extends in the direction orthogonal to the fitting axis and on which the connection object is placed, and the support surface is constituted of a bottom surface of a recessed shape portion formed at the placement surface.

8. The connector according to claim 1, comprising a top insulator being provided with a contact through-hole through which the tubular portion of the plug contact passes and which is smaller than the flange, wherein the top insulator is fixed to the bottom insulator such that the tubular portion of the plug contact passes through the contact through-hole and that the flange of the plug contact, the connection object, and the elastic portion of the inner contact are sandwiched between the top insulator and the bottom insulator.